

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	Uses the common ratio $\frac{5+2\sin\theta}{12\cos\theta} = \frac{6\tan\theta}{5+2\sin\theta}$ o.e.	M1	3.1a
	Cross multiplies and uses $\tan\theta \times \cos\theta = \sin\theta$ $(5+2\sin\theta)^2 = 6 \times 12 \sin\theta$	dM1	1.1b
	Proceeds to given answer $25 + 20\sin\theta + 4\sin^2\theta = 72\sin\theta$ $\Rightarrow 4\sin^2\theta - 52\sin\theta + 25 = 0$ *	A1*	2.1
		(3)	
(a) Alt	(a) Alternative example:		
	Uses the common ratio $12r \cos\theta = 5 + 2\sin\theta$, $12r^2 \cos\theta = 6\tan\theta$ $\Rightarrow 12\cos\theta \left(\frac{5+2\sin\theta}{12\cos\theta} \right)^2 = 6\tan\theta$	M1	3.1a
	Multiplies up and uses $\tan\theta \times \cos\theta = \sin\theta$ $(5+2\sin\theta)^2 = 6\tan\theta \times 12\cos\theta = 72\sin\theta$	dM1	1.1b
	Proceeds to given answer $25 + 20\sin\theta + 4\sin^2\theta = 72\sin\theta$ $\Rightarrow 4\sin^2\theta - 52\sin\theta + 25 = 0$ *	A1*	2.1
		(3)	
(b)	$4\sin^2\theta - 52\sin\theta + 25 = 0 \Rightarrow \sin\theta = \frac{1}{2} \left(\frac{25}{2} \right)$	M1	1.1b
	$\theta = \frac{5\pi}{6}$	A1	1.2
		(2)	

(c)	Attempts a value for either a or r e.g. $a = 12\cos\theta = 12 \times -\frac{\sqrt{3}}{2}$ or $r = \frac{5+2\sin\theta}{12\cos\theta} = \frac{5+2 \times \frac{1}{2}}{12 \times -\frac{\sqrt{3}}{2}}$	M1	3.1a
	" a " = $-6\sqrt{3}$ and " r " = $-\frac{1}{\sqrt{3}}$ o.e.	A1	1.1b
	Uses $S_{\infty} = \frac{a}{1-r} = \frac{-6\sqrt{3}}{1+\frac{1}{\sqrt{3}}}$	dM1	2.1
	Rationalises denominator $S_{\infty} = \frac{-6\sqrt{3}}{1+\frac{1}{\sqrt{3}}} = \frac{-18}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$	ddM1	1.1b
	$(S_{\infty})9(1-\sqrt{3})$	A1	2.1
		(5)	
(10 marks)			
Notes:			

(a)

M1: For the key step in using the ratio of $\frac{a_2}{a_1} = \frac{a_3}{a_2}$

dM1: Cross multiplies and uses $\tan\theta \times \cos\theta = \sin\theta$

A1*: Proceeds to the given answer including the " $= 0$ " with no errors and sufficient working shown.

Alternative:

M1: Expresses the 2nd and 3rd terms in terms of the first term and the common ratio and eliminates "r"

dM1: Multiplies up and uses $\tan \theta \times \cos \theta = \sin \theta$

A1*: Proceeds to the given answer including the " $= 0$ " with no errors and sufficient working shown.

Other approaches may be seen in (a) and can be marked in a similar way e.g. M1 for correctly obtaining an equation in θ using the GP, M1 for applying $\tan \theta \times \cos \theta = \sin \theta$ or equivalent and eliminating fractions, A1 as above

Example: $u_2 = \frac{u_1 \times u_3}{u_2} \Rightarrow 5 + 2 \sin \theta = \frac{12 \cos \theta \times 6 \tan \theta}{5 + 2 \sin \theta}$ **M1**

$\Rightarrow (5 + 2 \sin \theta)^2 = 72 \sin \theta$ **dM1**

$25 + 20 \sin \theta + 4 \sin^2 \theta = 72 \sin \theta$ **A1**

$\Rightarrow 4 \sin^2 \theta - 52 \sin \theta + 25 = 0$ *

(b)

M1: Attempts to solve $4 \sin^2 \theta - 52 \sin \theta + 25 = 0$. Must be clear they have found $\sin \theta$ and not e.g. just x from $4x^2 - 52x + 25 = 0$. Working does not need to be seen but see general guidance for solving a 3TQ if necessary. Note that the $\frac{25}{2}$ does not need to be seen.

A1: $\theta = \frac{5\pi}{6}$ and no other values unless they are rejected or the $\frac{5\pi}{6}$ clearly selected here and not in (c)

A minimum requirement in (b) is e.g. $\sin \theta = \frac{1}{2}$, $\theta = \frac{5\pi}{6}$

Do not allow 150° for $\frac{5\pi}{6}$

(c) Allow full marks in (c) if e.g. $\theta = \frac{\pi}{6}$ is their answer to (b) but $\theta = \frac{5\pi}{6}$ is used here.

or if e.g. $\theta = \frac{5\pi}{6}$ is their answer to (b) but $\theta = \frac{\pi}{6}$ is used here allow the M marks only.

M1: For attempting a value (exact or decimal) for either a or r using their θ

E.g. $a = 12 \cos \theta = \left(12 \times -\frac{\sqrt{3}}{2}\right)$ or $r = \frac{5 + 2 \sin \theta}{12 \cos \theta} = \left(\frac{5 + 2 \times \frac{1}{2}}{12 \times -\frac{\sqrt{3}}{2}}\right)$ oe e.g. $r = \frac{6 \tan \theta}{5 + 2 \sin \theta} = \left(\frac{6 \times -\frac{1}{\sqrt{3}}}{5 + 2 \times \frac{1}{2}}\right)$

A1: Finds both $a = -6\sqrt{3}$ and $r = -\frac{1}{\sqrt{3}}$ which can be left unsimplified but $\sin \theta = \frac{1}{2}$, $\cos \theta = -\frac{\sqrt{3}}{2}$

and $\tan \theta = -\frac{\sqrt{3}}{3}$ (if required) must have been used.

dM1: Uses both values of "a" and "r" with the equation $S_\infty = \frac{a}{1-r} = \frac{-6\sqrt{3}}{1+\frac{1}{\sqrt{3}}}$ to create an expression

involving surds where a and r have come from appropriate work and $|r| < 1$

Depends on the first method mark.

ddM1: Rationalises denominator. The denominator must be of the form $p \pm q\sqrt{3}$ oe e.g. $p + \frac{q}{\sqrt{3}}$

Depends on both previous method marks.

Note that stating e.g. $\frac{k}{p+q\sqrt{3}} \times \frac{p-q\sqrt{3}}{p-q\sqrt{3}}$ or $\frac{k}{p+\frac{q}{\sqrt{3}}} \times \frac{p-\frac{q}{\sqrt{3}}}{p-\frac{q}{\sqrt{3}}}$ is sufficient.

A1: Obtains $(S_{\infty} =) 9(1 - \sqrt{3})$

Note that full marks are available in (c) for the use of $\theta = 150^\circ$

Note also that marks may be implied in (c) by e.g.

$$\begin{aligned}
 S_{\infty} &= \frac{a}{1-r} = \frac{12 \cos \theta}{1 - \frac{5+2 \sin \theta}{12 \cos \theta}} = \frac{144 \cos^2 \theta}{12 \cos \theta - 5 - 2 \sin \theta} = \frac{144 \cos^2 \frac{5\pi}{6}}{12 \cos \frac{5\pi}{6} - 5 - 2 \sin \frac{5\pi}{6}} \\
 &= \frac{108}{-6 - 6\sqrt{3}} = \frac{108}{-6 - 6\sqrt{3}} \times \frac{-6 + 6\sqrt{3}}{-6 + 6\sqrt{3}} = \frac{-648 + 648\sqrt{3}}{-72} = 9(1 - \sqrt{3})
 \end{aligned}$$

Scores M1A1 implied dM1 ddM1 A1

$$S_{\infty} = \frac{a}{1-r} = \frac{12 \cos \frac{5\pi}{6}}{1 - \frac{5+2 \sin \frac{5\pi}{6}}{12 \cos \frac{5\pi}{6}}} \quad \text{or e.g.} \quad S_{\infty} = \frac{a}{1-r} = \frac{12 \cos \frac{\pi}{6}}{1 - \frac{5+2 \sin \frac{\pi}{6}}{12 \cos \frac{\pi}{6}}}$$

And nothing else

scores M1A0dM1ddM0A0

$$S_{\infty} = \frac{a}{1-r} = \frac{12 \cos \frac{5\pi}{6}}{1 - \frac{5+2 \sin \frac{5\pi}{6}}{12 \cos \frac{5\pi}{6}}} = 9(1 - \sqrt{3})$$

Scores M1A1dM1ddM0A0

$$S_{\infty} = \frac{a}{1-r} = \frac{12 \cos \frac{\pi}{6}}{1 - \frac{5+2 \sin \frac{\pi}{6}}{12 \cos \frac{\pi}{6}}} = 9(1 + \sqrt{3})$$

Scores M1A0dM1ddM0A0

$S_{\infty} = 9(1 - \sqrt{3})$ with no working scores no marks

Q2.

Question	Scheme	Marks	AOs
	Use of $\frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta}$	B1	2.1
	Uses the compound angle identity for $\sin(A+B)$ with $A = \theta$, $B = h$ $\Rightarrow \sin(\theta+h) = \sin \theta \cos h + \cos \theta \sin h$	M1	1.1b
	Achieves $\frac{\sin(\theta+h) - \sin \theta}{h} = \frac{\sin \theta \cos h + \cos \theta \sin h - \sin \theta}{h}$	A1	1.1b
	$= \frac{\sin h}{h} \cos \theta + \left(\frac{\cos h - 1}{h} \right) \sin \theta$	M1	2.1
	Uses $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$ Hence the $\lim_{h \rightarrow 0} \frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta} = \cos \theta$ and the gradient of the chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos \theta$ *	A1*	2.5
(5 marks)			

Notes:

B1: States or implies that the gradient of the chord is $\frac{\sin(\theta+h) - \sin \theta}{h}$ or similar such as

$$\frac{\sin(\theta+\delta\theta) - \sin \theta}{\theta + \delta\theta - \theta} \text{ for a small } h \text{ or } \delta\theta$$

M1: Uses the compound angle identity for $\sin(A+B)$ with $A = \theta$, $B = h$ or $\delta\theta$

A1: Obtains $\frac{\sin \theta \cos h + \cos \theta \sin h - \sin \theta}{h}$ or equivalent

M1: Writes their expression in terms of $\frac{\sin h}{h}$ and $\frac{\cos h - 1}{h}$

A1*: Uses correct language to explain that $\frac{dy}{d\theta} = \cos \theta$

For this method they should use all of the given statements $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$,

$$\frac{\cos h - 1}{h} \rightarrow 0 \text{ meaning that the } \lim_{h \rightarrow 0} \frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta} = \cos \theta$$

and therefore the gradient of the chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos \theta$

Question	Scheme	Marks	AOs
alt	Use of $\frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta}$	B1	2.1
	Sets $\frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta} = \frac{\sin\left(\theta + \frac{h}{2} + \frac{h}{2}\right) - \sin\left(\theta + \frac{h}{2} - \frac{h}{2}\right)}{h}$ and uses the compound angle identity for $\sin(A+B)$ and $\sin(A-B)$ with $A = \theta + \frac{h}{2}$, $B = \frac{h}{2}$	M1	1.1b
	Achieves $\frac{\sin(\theta+h) - \sin \theta}{h} = \frac{\left[\sin\left(\theta + \frac{h}{2}\right)\cos\left(\frac{h}{2}\right) + \cos\left(\theta + \frac{h}{2}\right)\sin\left(\frac{h}{2}\right)\right] - \left[\sin\left(\theta + \frac{h}{2}\right)\cos\left(\frac{h}{2}\right) - \cos\left(\theta + \frac{h}{2}\right)\sin\left(\frac{h}{2}\right)\right]}{h}$	A1	1.1b
	$= \frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \times \cos\left(\theta + \frac{h}{2}\right)$	M1	2.1
	Uses $h \rightarrow 0, \frac{h}{2} \rightarrow 0$ hence $\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \rightarrow 1$ and $\cos\left(\theta + \frac{h}{2}\right) \rightarrow \cos \theta$ Therefore the $\lim_{h \rightarrow 0} \frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta} = \cos \theta$ and the gradient of the chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos \theta$ *	A1*	2.5
(5 marks)			

Additional notes:

A1*: Uses correct language to explain that $\frac{dy}{d\theta} = \cos \theta$. For this method they should use the

(adapted) given statement $h \rightarrow 0, \frac{h}{2} \rightarrow 0$ hence $\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \rightarrow 1$ with $\cos\left(\theta + \frac{h}{2}\right) \rightarrow \cos \theta$

meaning that the $\lim_{h \rightarrow 0} \frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta} = \cos \theta$ and therefore the gradient of the

chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos \theta$

Q3.

Question	Scheme	Marks	AOs
(a)	Attempts $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$	M1	1.1b
	$\frac{dy}{dx} = \frac{\sqrt{3} \sin 2t}{\sin t} \quad (= 2\sqrt{3} \cos t)$	A1	1.1b
	(2)		
(b)	Substitutes $t = \frac{2\pi}{3}$ in $\frac{dy}{dx} = \frac{\sqrt{3} \sin 2t}{\sin t} = (-\sqrt{3})$	M1	2.1
	Uses gradient of normal $= -\frac{1}{dy/dx} = \left(\frac{1}{\sqrt{3}}\right)$	M1	2.1
	Coordinates of P $= \left(-1, -\frac{\sqrt{3}}{2}\right)$	B1	1.1b
	Correct form of normal $y + \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}(x+1)$	M1	2.1
	Completes proof $\Rightarrow 2x - 2\sqrt{3}y - 1 = 0$ *	A1*	1.1b
	(5)		
(c)	Substitutes $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$ into $2x - 2\sqrt{3}y - 1 = 0$	M1	3.1a
	Uses the identity $\cos 2t = 2\cos^2 t - 1$ to produce a quadratic in $\cos t$	M1	3.1a
	$\Rightarrow 12\cos^2 t - 4\cos t - 5 = 0$	A1	1.1b
	Finds $\cos t = \frac{5}{6}, \frac{1}{2}$	M1	2.4
	Substitutes their $\cos t = \frac{5}{6}$ into $x = 2\cos t$, $y = \sqrt{3} \cos 2t$,	M1	1.1b
	$Q = \left(\frac{5}{3}, \frac{7}{18}\sqrt{3}\right)$	A1	1.1b
	(6)		
(13 marks)			

Notes:

(a)

M1: Attempts $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ and achieves a form $k \frac{\sin 2t}{\sin t}$ Alternatively candidates may apply the double angle identity for $\cos 2t$ and achieve a form $k \frac{\sin t \cos t}{\sin t}$

A1: Scored for a correct answer, either $\frac{\sqrt{3} \sin 2t}{\sin t}$ or $2\sqrt{3} \cos t$

(b)

M1: For substituting $t = \frac{2\pi}{3}$ in their $\frac{dy}{dx}$ which must be in terms of t

M1: Uses the gradient of the normal is the negative reciprocal of the value of $\frac{dy}{dx}$. This may be seen in the equation of l .

B1: States or uses (in their tangent or normal) that $P = \left(-1, -\frac{\sqrt{3}}{2}\right)$

M1: Uses their numerical value of $-1/\frac{dy}{dx}$ with their $\left(-1, -\frac{\sqrt{3}}{2}\right)$ to form an equation of the normal at P

A1*: This is a proof and all aspects need to be correct. Correct answer only $2x - 2\sqrt{3}y - 1 = 0$

(c)

M1: For substituting $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$ into $2x - 2\sqrt{3}y - 1 = 0$ to produce an equation in t . Alternatively candidates could use $\cos 2t = 2\cos^2 t - 1$ to set up an equation of the form $y = Ax^2 + B$.

M1: Uses the identity $\cos 2t = 2\cos^2 t - 1$ to produce a quadratic equation in $\cos t$
In the alternative method it is for combining their $y = Ax^2 + B$ with $2x - 2\sqrt{3}y - 1 = 0$ to get an equation in just one variable

A1: For the correct quadratic equation $12\cos^2 t - 4\cos t - 5 = 0$

Alternatively the equations in x and y are $3x^2 - 2x - 5 = 0$ $12\sqrt{3}y^2 + 4y - 7\sqrt{3} = 0$

M1: Solves the quadratic equation in $\cos t$ (or x or y) and rejects the value corresponding to P .

M1: Substitutes their $\cos t = \frac{5}{6}$ or their $t = \arccos\left(\frac{5}{6}\right)$ in $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$

If a value of x or y has been found it is for finding the other coordinate.

A1: $Q = \left(\frac{5}{3}, \frac{7}{18}\sqrt{3}\right)$. Allow $x = \frac{5}{3}, y = \frac{7}{18}\sqrt{3}$ but do not allow decimal equivalents.

Q4.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{dx}{dy} = 8 \cos 2y \Rightarrow \frac{dy}{dx} = \frac{1}{8 \cos 2y}$	M1	This mark is given for differentiating and inverting
	At (0, 0), $\frac{dy}{dx} = \frac{1}{8}$	A1	This mark is given for finding $\frac{dy}{dx}$ when $y = 0$
(b)(i)	$\sin 2y \approx 2y \Rightarrow x \approx 8y$	B1	This mark is given for finding an approximation for x
(b)(ii)	When x and y are small, $x = 4 \sin 2y$ approximates to the line $x = 8y$	B1	This mark is given for a valid explanation of the relationship between x and y when both are small
(c)	$\sin^2 2y + \cos^2 2y = 1$ $\Rightarrow \cos^2 2y = 1 - \sin^2 2y$ $x = 4 \sin 2y \Rightarrow \sin^2 2y = \left(\frac{x}{4}\right)^2$	M1	This mark is given for a method to use find an expression for $\sin^2 2y$ in terms of x
	$\frac{dy}{dx} = \frac{1}{8 \cos 2y} = \frac{1}{8 \sqrt{1 - \left(\frac{x}{4}\right)^2}}$	A1	This mark is given for an unsimplified expression for $\frac{dy}{dx}$
	$\frac{dy}{dx} = \frac{1}{2\sqrt{16 - x^2}}$	A1	This mark is given for a fully correct answer with $a = 2$ and $b = 16$

Q5.

Question	Scheme	Marks	AOs
(a)	$\tan \theta + \cot \theta \equiv \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$	M1	2.1
	$\equiv \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$	A1	1.1b
	$\equiv \frac{1}{\frac{1}{2} \sin 2\theta}$	M1	2.1
	$\equiv 2 \operatorname{cosec} 2\theta$ *	A1*	1.1b
		(4)	
(b)	States $\tan \theta + \cot \theta = 1 \Rightarrow \sin 2\theta = 2$ AND no real solutions as $-1 \leq \sin 2\theta \leq 1$	B1	2.4
		(1)	
(5 marks)			

Notes:

(a)

M1: Writes $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and $\cot \theta = \frac{\cos \theta}{\sin \theta}$

A1: Achieves a correct intermediate answer of $\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$

M1: Uses the double angle formula $\sin 2\theta = 2 \sin \theta \cos \theta$

A1*: Completes proof with no errors. This is a given answer.

Note: There are many alternative methods. For example

$$\tan \theta + \cot \theta \equiv \tan \theta + \frac{1}{\tan \theta} \equiv \frac{\tan^2 \theta + 1}{\tan \theta} \equiv \frac{\sec^2 \theta}{\tan \theta} \equiv \frac{1}{\cos^2 \theta \times \frac{\sin \theta}{\cos \theta}} \equiv \frac{1}{\cos \theta \times \sin \theta} \text{ then as the}$$

main scheme.

(b)

B1: Scored for sight of $\sin 2\theta = 2$ and a reason as to why this equation has no real solutions. Possible reasons could be $-1 \leq \sin 2\theta \leq 1$and therefore $\sin 2\theta \neq 2$ or $\sin 2\theta = 2 \Rightarrow 2\theta = \arcsin 2$ which has no answers as $-1 \leq \sin 2\theta \leq 1$

Q6.

Question	Scheme	Marks	AOs
(a)	$D = 5 + 2 \sin(30 \times 6.5)^\circ = \text{awrt } 4.48 \text{ m with units}$	B1	3.4
		(1)	
(b)	$3.8 = 5 + 2 \sin(30t)^\circ \Rightarrow \sin(30t)^\circ = -0.6$	M1	1.1b
		A1	1.1b
	$t = 10.77$	dM1	3.1a
	10:46 a.m. or 10:47 a.m.	A1	3.2a
		(4)	
(5 marks)			

Notes:

(a)

B1: Scored for using the model ie. substituting $t = 6.5$ into $D = 5 + 2\sin(30t)^\circ$ and stating $D = \text{awrt } 4.48\text{m}$. The units must be seen somewhere in (a). So allow when $D = 4.482... = 4.5\text{ m}$
Allow the mark for a correct answer without any working.

(b)

M1: For using $D = 3.8$ and proceeding to $\sin(30t)^\circ = k$, $|k| \leq 1$

A1: $\sin(30t)^\circ = -0.6$ This may be implied by any correct answer for t such as $t = 7.2$

If the A1 implied, the calculation must be performed in degrees.

dM1: For finding the first value of t for their $\sin(30t)^\circ = k$ after $t = 8.5$.

You may well see other values as well which is not an issue for this dM mark

(Note that $\sin(30t)^\circ = -0.6 \Rightarrow 30t = 216.9^\circ$ as well but this gives $t = 7.2$)

For the correct $\sin(30t)^\circ = -0.6 \Rightarrow 30t = 323.1 \Rightarrow t = \text{awrt } 10.8$

For the incorrect $\sin(30t)^\circ = +0.6 \Rightarrow 30t = 396.9 \Rightarrow t = \text{awrt } 13.2$

So award this mark if you see $30t = \text{inv sin their } -0.6$ to give the first value of t where $30t > 255$

A1: Allow 10:46 a.m. (12 hour clock notation) or 10:46 (24 hour clock notation) oe

Allow 10:47 a.m. (12 hour clock notation) or 10:47 (24 hour clock notation) oe

DO NOT allow 646 minutes or 10 hours 46 minutes.

Q7.

Question	Scheme	Marks	AOs
(a)	Identifies an error for student A: They use $\frac{\cos \theta}{\sin \theta} = \tan \theta$ It should be $\frac{\sin \theta}{\cos \theta} = \tan \theta$	B1	2.3
		(1)	
(b)	(i) Shows $\cos(-26.6^\circ) \neq 2 \sin(-26.6^\circ)$, so cannot be a solution	B1	2.4
	(ii) Explains that the incorrect answer was introduced by squaring	B1	2.4
		(2)	
(3 marks)			
Notes:			
(a)	B1: Accept a response of the type 'They use $\frac{\cos \theta}{\sin \theta} = \tan \theta$. This is incorrect as $\frac{\sin \theta}{\cos \theta} = \tan \theta$ ' It can be implied by a response such as 'They should get $\tan \theta = \frac{1}{2}$ not $\tan \theta = 2$ ' Accept also statements such as 'it should be $\cot \theta = 2$ '		
(b)	B1: Accept a response where the candidate shows that -26.6° is not a solution of $\cos \theta = 2 \sin \theta$. This can be shown by, for example, finding both $\cos(-26.6^\circ)$ and $2 \sin(-26.6^\circ)$ and stating that they are not equal. An acceptable alternative is to state that $\cos(-26.6^\circ) = +ve$ and $2 \sin(-26.6^\circ) = -ve$ and stating that they therefore cannot be equal. B1: Explains that the incorrect answer was introduced by squaring. Accept an example showing this. For example $x = 5$ squared gives $x^2 = 25$ which has answers ± 5		

Q8.

Question	Scheme	Marks	AOs
(a)	$\cos 3A = \cos(2A + A) = \cos 2A \cos A - \sin 2A \sin A$	M1	3.1a
	$= (2 \cos^2 A - 1) \cos A - (2 \sin A \cos A) \sin A$	dM1	1.1b
	$= (2 \cos^2 A - 1) \cos A - 2 \cos A (1 - \cos^2 A)$	ddM1	2.1

	$= 4\cos^3 A - 3\cos A^*$	A1*	1.1b
		(4)	
(b)	$1 - \cos 3x = \sin^2 x \Rightarrow \cos^2 x + 3\cos x - 4\cos^3 x = 0$	M1	1.1b
	$\Rightarrow \cos x(4\cos^2 x - \cos x - 3) = 0$ $\Rightarrow \cos x(4\cos x + 3)(\cos x - 1) = 0$ $\Rightarrow \cos x = \dots$	dM1	3.1a
	Two of $-90^\circ, 0, 90^\circ$, awrt 139°	A1	1.1b
	All four of $-90^\circ, 0, 90^\circ$, awrt 139°	A1	2.1
		(4)	
(8 marks)			

Notes:

(a)

Allow a proof in terms of x rather than A

M1: Attempts to use the compound angle formula for $\cos(2A + A)$ or $\cos(A + 2A)$

Condone a slip in sign

dM1: Uses correct double angle identities for $\cos 2A$ and $\sin 2A$

$\cos 2A = 2\cos^2 A - 1$ must be used. If either of the other two versions are used expect to see an attempt to replace $\sin^2 A$ by $1 - \cos^2 A$ at a later stage.

Depends on previous mark.

ddM1: Attempts to get all terms in terms of $\cos A$ using correct and appropriate identities.

Depends on both previous marks.

A1*: A completely correct and rigorous proof including correct notation, no mixed variables, missing brackets etc.

Alternative right to left is possible:

$$4\cos^3 A - 3\cos A = \cos A(4\cos^2 A - 3) = \cos A(2\cos^2 A - 1 + 2(1 - \sin^2 A) - 2) = \cos A(\cos 2A - 2\sin^2 A)$$

$$= \cos A \cos 2A - 2\sin A \cos A \sin A = \cos A \cos 2A - \sin 2A \sin A = \cos(2A + A) = \cos 3A$$

Score M1: For $4\cos^3 A - 3\cos A = \cos A(4\cos^2 A - 3)$

dM1: For $\cos A(2\cos^2 A - 1 + 2(1 - \sin^2 A) - 2)$ (Replaces $4\cos^2 A - 1$ by $2\cos^2 A - 1$ and $2(1 - \sin^2 A)$)

ddM1: Reaches $\cos A \cos 2A - \sin 2A \sin A$

A1: $\cos(2A + A) = \cos 3A$

(b)

M1: For an attempt to produce an equation just in $\cos x$ using both part (a) and the identity $\sin^2 x = 1 - \cos^2 x$

Allow one slip in sign or coefficient when copying the result from part (a)

dM1: Dependent upon the preceding mark. It is for taking the cubic equation in $\cos x$ and making a valid attempt to solve. This could include factorisation or division of a $\cos x$ term followed by an attempt to solve the 3 term quadratic equation in $\cos x$ to reach at least one non zero value for $\cos x$.

May also be scored for solving the cubic equation in $\cos x$ to reach at least one non zero value for $\cos x$.

A1: Two of $-90^\circ, 0, 90^\circ$, awrt 139° **Depends on the first method mark.**

A1: All four of $-90^\circ, 0, 90^\circ$, awrt 139° with no extra solutions offered within the range.

Note that this is an alternative approach for obtaining the cubic equation in (b):

$$1 - \cos 3x = \sin^2 x \Rightarrow 1 - \cos 3x = \frac{1}{2}(1 - \cos 2x)$$

$$\Rightarrow 2 - 2\cos 3x = 1 - \cos 2x$$

$$\Rightarrow 1 = 2\cos 3x - \cos 2x$$

$$\Rightarrow 1 = 2(4\cos^3 x - 3\cos x) - (2\cos^2 x - 1)$$

$$\Rightarrow 0 = 4\cos^3 x - 3\cos x - \cos^2 x$$

The M1 will be scored on the penultimate line when they use part (a) and use the correct identity for $\cos 2x$

Q9.

Question	Scheme	Marks	AOs
(a)	$R = \sqrt{109}$	B1	1.1b
	$\tan \alpha = \frac{3}{10}$	M1	1.1b
	$\alpha = 16.70^\circ$ so $\sqrt{109} \cos(\theta + 16.70^\circ)$	A1	1.1b
		(3)	
(b)	(i) e.g. $H = 11 - 10 \cos(80t)^\circ + 3 \sin(80t)^\circ$ or $H = 11 - \sqrt{109} \cos(80t + 16.70)^\circ$	B1	3.3
	(ii) $11 + \sqrt{109}$ or 21.44 m	B1ft	3.4
		(2)	
(c)	Sets $80t + "16.70" = 540$	M1	3.4
	$t = \frac{540 - "16.70"}{80} = (6.54)$	M1	1.1b
	$t = 6$ mins 32 seconds	A1	1.1b
		(3)	
(d)	Increase the '80' in the formula For example use $H = 11 - 10 \cos(90t)^\circ + 3 \sin(90t)^\circ$		3.3
		(1)	
(9 marks)			

Notes:
(a) B1: $R = \sqrt{109}$ Do not allow decimal equivalents M1: Allow for $\tan \alpha = \pm \frac{3}{10}$ A1: $\alpha = 16.70^\circ$
(b)(i) B1: see scheme (b)(ii) B1ft: their $11 +$ their $\sqrt{109}$ Allow decimals here.
(c) M1: Sets $80t + "16.70" = 540$. Follow through on their 16.70 M1: Solves their $80t + "16.70" = 540$ correctly to find t A1: $t = 6$ mins 32 seconds
(d) B1: States that to increase the speed of the wheel the 80's in the equation would need to be increased.

Q10.

Part	Working or answer an examiner might expect to see	Mark	Notes
	$(10 \cos t)^2 + (4\sqrt{2} \sin t)^2 = 66$	M1	This mark is given for combining the two equations to show where the curve and circle meet
	$100 (\cos t)^2 + 32(1 - \cos t)^2 = 66$	M1	This mark is given for forming an equation in $\cos t$ only
	$68 \cos^2 t = 34$	A1	This mark is given for simplifying to find an equation in terms of $\cos t$
	$\cos t = \pm \frac{1}{\sqrt{2}} \Rightarrow t = \frac{\pi}{4}$	M1	This mark is given for finding a value for t
	$x = 10 \times \frac{1}{\sqrt{2}}$ $y = 4\sqrt{2} \times -\sin \frac{\pi}{4} = 4\sqrt{2} \times -\frac{1}{\sqrt{2}}$	M1	This mark is given for a method to substitute back into the original equations to find value for x and y
	$S = (5\sqrt{2}, -4)$	A1	This mark is given for the correct coordinates of S
(Total 6 marks)			

Q11.

Question	Scheme	Marks	AOs
	$1 - \cos 2\theta \equiv \tan \theta \sin 2\theta, \theta \neq \frac{(2n+1)\pi}{2}, n \in \mathbb{Z}$		
(a) Way 1	$\tan \theta \sin 2\theta = \left(\frac{\sin \theta}{\cos \theta}\right)(2 \sin \theta \cos \theta)$	M1	1.1b
	$= \left(\frac{\sin \theta}{\cancel{\cos \theta}}\right)(2 \sin \theta \cancel{\cos \theta}) = 2 \sin^2 \theta = 1 - \cos 2\theta *$	M1	1.1b
		A1*	2.1
		(3)	
(a) Way 2	$1 - \cos 2\theta = 1 - (1 - 2 \sin^2 \theta) = 2 \sin^2 \theta$	M1	1.1b
	$= \left(\frac{\sin \theta}{\cos \theta}\right)(2 \sin \theta \cos \theta) = \tan \theta \sin 2\theta *$	M1	1.1b
		A1*	2.1
		(3)	
	$(\sec^2 x - 5)(1 - \cos 2x) = 3 \tan^2 x \sin 2x, -\frac{\pi}{2} < x < \frac{\pi}{2}$		
(b) Way 1	$(\sec^2 x - 5) \tan x \sin 2x = 3 \tan^2 x \sin 2x$ or $(\sec^2 x - 5)(1 - \cos 2x) = 3 \tan x(1 - \cos 2x)$		
	Deduces $x = 0$	B1	2.2a
	Uses $\sec^2 x = 1 + \tan^2 x$ and cancels/factorises out $\tan x$ or $(1 - \cos 2x)$ e.g. $(1 + \tan^2 x - 3 \tan x - 5) \tan x = 0$ or $(1 + \tan^2 x - 3 \tan x - 5)(1 - \cos 2x) = 0$ or $1 + \tan^2 x - 5 = 3 \tan x$	M1	2.1
	$\tan^2 x - 3 \tan x - 4 = 0$	A1	1.1b
	$(\tan x - 4)(\tan x + 1) = 0 \Rightarrow \tan x = \dots$	M1	1.1b
	$x = -\frac{\pi}{4}, 1.326$	A1	1.1b
		A1	1.1b
		(6)	

(9 marks)

Notes for Question	
(a)	Way 1
M1:	Applies $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and $\sin 2\theta = 2 \sin \theta \cos \theta$ to $\tan \theta \sin 2\theta$
M1:	Cancels as scheme (may be implied) and attempts to use $\cos 2\theta = 1 - 2 \sin^2 \theta$
A1*:	For a correct proof showing all steps of the argument
(a)	Way 2
M1:	For using $\cos 2\theta = 1 - 2 \sin^2 \theta$
Note:	If the form $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ or $\cos 2\theta = 2 \cos^2 \theta - 1$ is used, the mark cannot be awarded until $\cos^2 \theta$ has been replaced by $1 - \sin^2 \theta$
M1:	Attempts to write their $2 \sin^2 \theta$ in terms of $\tan \theta$ and $\sin 2\theta$ using $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and $\sin 2\theta = 2 \sin \theta \cos \theta$ within the given expression
A1*:	For a correct proof showing all steps of the argument
Note:	If a proof meets in the middle; e.g. they show $\text{LHS} = 2 \sin^2 \theta$ and $\text{RHS} = 2 \sin^2 \theta$; then some indication must be given that the proof is complete. E.g. $1 - \cos 2\theta \equiv \tan \theta \sin 2\theta$, QED, box

Notes for Question Continued			
(b)			
B1:	Deduces that the given equation yields a solution $x = 0$		
M1:	For using the key step of $\sec^2 x = 1 + \tan^2 x$ and cancels/factorises out $\tan x$ or $(1 - \cos 2x)$ or $\sin 2x$ to produce a quadratic factor or quadratic equation in just $\tan x$		
Note:	Allow the use of $\pm \sec^2 x = \pm 1 \pm \tan^2 x$ for M1		
A1:	Correct 3TQ in $\tan x$. E.g. $\tan^2 x - 3 \tan x - 4 = 0$		
Note:	E.g. $\tan^2 x - 4 = 3 \tan x$ or $\tan^2 x - 3 \tan x = 4$ are acceptable for A1		
M1:	For a correct method of solving their 3TQ in $\tan x$		
A1:	Any one of $-\frac{\pi}{4}$, awrt -0.785 , awrt 1.326 , -45° , awrt 75.964°		
A1:	Only $x = -\frac{\pi}{4}, 1.326$ cao stated in the range $-\frac{\pi}{2} < x < \frac{\pi}{2}$		
Note:	Alternative Method (Alt 1)		
	$(\sec^2 x - 5) \tan x \sin 2x = 3 \tan^2 x \sin 2x$ or $(\sec^2 x - 5)(1 - \cos 2x) = 3 \tan x(1 - \cos 2x)$		
	Deduces $x = 0$		B1 2.2a
	$\sec^2 x - 5 = 3 \tan x \Rightarrow \frac{1}{\cos^2 x} - 5 = 3 \left(\frac{\sin x}{\cos x} \right)$ $1 - 5 \cos^2 x = 3 \sin x \cos x$ $1 - 5 \left(\frac{1 + \cos 2x}{2} \right) = \frac{3}{2} \sin 2x$ $-\frac{3}{2} - \frac{5}{2} \cos 2x = \frac{3}{2} \sin 2x$ $\{3 \sin 2x + 5 \cos 2x = -3\}$	Complete process (as shown) of using the identities for $\sin 2x$ and $\cos 2x$ to proceed as far as $\pm A \pm B \cos 2x = \pm C \sin 2x$ $-\frac{3}{2} - \frac{5}{2} \cos 2x = \frac{3}{2} \sin 2x$ o.e.	M1 2.1
	$\sqrt{34} \sin(2x + 1.03) = -3$	Expresses their answer in the form $R \sin(2x + \alpha) = k$; $k \neq 0$ with values for R and α	A1 1.1b
	$\sin(2x + 1.03) = -\frac{3}{\sqrt{34}}$		
	$x = -\frac{\pi}{4}, 1.326$		A1 1.1b
			A1 1.1b

Q12.

Question	Scheme	Marks	AOs
	$\frac{d}{d\theta}(\cos \theta) = -\sin \theta$; as $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$		
	$\frac{\cos(\theta + h) - \cos \theta}{h}$	B1	2.1
	$= \frac{\cos \theta \cos h - \sin \theta \sin h - \cos \theta}{h}$	M1	1.1b
		A1	1.1b
	$= -\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta$		
	As $h \rightarrow 0$, $-\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta \rightarrow -1 \sin \theta + 0 \cos \theta$	dM1	2.1
	so $\frac{d}{d\theta}(\cos \theta) = -\sin \theta$ *	A1*	2.5
		(5)	
(5 marks)			

Notes for Question	
B1:	Gives the correct fraction such as $\frac{\cos(\theta + h) - \cos \theta}{h}$ or $\frac{\cos(\theta + \delta\theta) - \cos \theta}{\delta\theta}$ Allow $\frac{\cos(\theta + h) - \cos \theta}{(\theta + h) - \theta}$ o.e. Note: $\cos(\theta + h)$ or $\cos(\theta + \delta\theta)$ may be expanded
M1:	Uses the compound angle formula for $\cos(\theta + h)$ to give $\cos \theta \cos h \pm \sin \theta \sin h$
A1:	Achieves $\frac{\cos \theta \cos h - \sin \theta \sin h - \cos \theta}{h}$ or equivalent
dM1:	dependent on both the B and M marks being awarded Complete attempt to apply the given limits to the gradient of their chord
Note:	They must isolate $\frac{\sin h}{h}$ and $\left(\frac{\cos h - 1}{h}\right)$, and replace $\frac{\sin h}{h}$ with 1 and replace $\left(\frac{\cos h - 1}{h}\right)$ with 0
A1*:	cso. Uses correct mathematical language of limiting arguments to prove $\frac{d}{d\theta}(\cos \theta) = -\sin \theta$
Note:	Acceptable responses for the final A mark include: $\frac{d}{d\theta}(\cos \theta) = \lim_{h \rightarrow 0} \left(-\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta \right) = -1 \sin \theta + 0 \cos \theta = -\sin \theta$ - Gradient of chord $= -\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta$. As $h \rightarrow 0$, gradient of chord tends to the gradient of the curve, so derivative is $-\sin \theta$ - Gradient of chord $= -\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta$. As $h \rightarrow 0$, gradient of curve is $-\sin \theta$
Note:	Give final A0 for the following example which shows <i>no limiting arguments</i> : when $h = 0$, $\frac{d}{d\theta}(\cos \theta) = -\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h}\right) \cos \theta = -1 \sin \theta + 0 \cos \theta = -\sin \theta$
Note:	Do not allow the final A1 for stating $\frac{\sin h}{h} = 1$ or $\left(\frac{\cos h - 1}{h}\right) = 0$ and attempting to apply these
Note:	In this question $\delta\theta$ may be used in place of h
Note:	Condone $f'(\theta)$ where $f(\theta) = \cos \theta$ or $\frac{dy}{d\theta}$ where $y = \cos \theta$ used in place of $\frac{d}{d\theta}(\cos \theta)$

Notes for Question Continued	
Note:	Condone x used in place of θ if this is done consistently
Note:	Give final A0 for $\frac{d}{d\theta}(\cos x) = \lim_{h \rightarrow 0} \left(-\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h} \right) \cos \theta \right) = -1 \sin \theta + 0 \cos \theta = -\sin \theta$ $\frac{d}{d\theta} = \dots$ - Defining $f(x) = \cos \theta$ and applying $f'(x) = \dots$ $\frac{d}{dx}(\cos \theta)$
Note:	Give final A1 for a correct limiting argument in x , followed by $\frac{d}{d\theta}(\cos \theta) = -\sin \theta$ e.g. $\frac{d}{d\theta}(\cos x) = \lim_{h \rightarrow 0} \left(-\frac{\sin h}{h} \sin x + \left(\frac{\cos h - 1}{h} \right) \cos x \right) = -1 \sin x + 0 \cos x = -\sin x$ $\Rightarrow \frac{d}{d\theta}(\cos \theta) = -\sin \theta$
Note:	Applying $h \rightarrow 0$, $\sin h \rightarrow h$, $\cos h \rightarrow 1$ to give e.g. $\lim_{h \rightarrow 0} \left(\frac{\cos \theta \cos h - \sin \theta \sin h - \cos \theta}{h} \right) = \left(\frac{\cos \theta(1) - \sin \theta(h) - \cos \theta}{h} \right) = \frac{-\sin \theta(h)}{h} = -\sin \theta$ is final M0 A0 for incorrect application of limits
Note:	$\lim_{h \rightarrow 0} \left(\frac{\cos \theta \cos h - \sin \theta \sin h - \cos \theta}{h} \right) = \lim_{h \rightarrow 0} \left(-\frac{\sin h}{h} \sin \theta + \left(\frac{\cos h - 1}{h} \right) \cos \theta \right)$ $= \lim_{h \rightarrow 0} (-1 \sin \theta + 0 \cos \theta) = -\sin \theta$. So for not removing $\lim_{h \rightarrow 0}$ when the limit was taken is final A0
Note:	Alternative Method: Considers $\frac{\cos(\theta+h) - \cos(\theta-h)}{(\theta+h) - (\theta-h)}$ which simplifies to $\frac{-2 \sin \theta \sin h}{2h}$

Q13.

Question	Scheme	Marks	AOs
(a)	Attempts to use both $\sin(x - 60^\circ) = \pm \sin x \cos 60^\circ \pm \cos x \sin 60^\circ$ $\cos(x - 30^\circ) = \pm \cos x \cos 30^\circ \pm \sin x \sin 30^\circ$	M1	2.1
	Correct equation $2 \sin x \cos 60^\circ - 2 \cos x \sin 60^\circ = \cos x \cos 30^\circ + \sin x \sin 30^\circ$	A1	1.1b
	Either uses $\frac{\sin x}{\cos x} = \tan x$ and attempts to make $\tan x$ the subject E.g. $(2 \cos 60^\circ - \sin 30^\circ) \tan x = \cos 30^\circ + 2 \sin 60^\circ$ Or attempts $\sin 30^\circ$ etc with at least two correct and collects terms in $\sin x$ and $\cos x$ E.g. $\left(2 \times \frac{1}{2} - \frac{1}{2}\right) \sin x = \left(2 \times \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\right) \cos x$	M1	2.1
	Proceeds to given answer showing all key steps E.g. $\frac{1}{2} \tan x = \frac{3\sqrt{3}}{2} \Rightarrow \tan x = 3\sqrt{3}$ *	A1*	1.1b
		(4)	
(b)	Deduces that $x = 2\theta + 60^\circ$	B1	2.2a
	$\tan(2\theta + 60^\circ) = 3\sqrt{3} \Rightarrow 2\theta + 60^\circ = 79.1^\circ, 259.1^\circ, \dots$	M1	1.1b
	Correct method to find one value of θ E.g. $\theta = \frac{79.1^\circ - 60^\circ}{2}$	dM1	1.1b
	$\theta = \text{awrt } 9.6^\circ, 99.6^\circ$ (See note)	A1	2.1
		(4)	
(8 marks)			
Notes:			

(a)

M1: Attempts to use both compound angle expansions to set up an equation in $\sin x$ and $\cos x$

The terms must be correct but condone sign errors and a slip on the multiplication of 2

A1: Correct equation $2 \sin x \cos 60^\circ - 2 \cos x \sin 60^\circ = \cos x \cos 30^\circ + \sin x \sin 30^\circ$ o.e.

Note that $\cos 60^\circ = \sin 30^\circ$ and $\cos 30^\circ = \sin 60^\circ$

Also allow this mark for candidates who substitute in their trigonometric values "early"

$$2 \sin x \times \frac{1}{2} - 2 \cos x \times \frac{\sqrt{3}}{2} = \cos x \times \frac{\sqrt{3}}{2} + \sin x \times \frac{1}{2} \quad \text{o.e.}$$

M1: Shows the necessary progress towards showing the given result.

There are three key moves, two of which must be shown for this mark.

- uses $\frac{\sin x}{\cos x} = \tan x$ to form an equation in just $\tan x$.
- uses exact numerical values for $\sin 30^\circ, \sin 60^\circ, \cos 30^\circ, \cos 60^\circ$ with at least two correct
- collects terms in $\sin x$ and $\cos x$ or alternatively in $\tan x$

A1*: Proceeds to the given answer with accurate work showing all necessary lines.

Examples of two proofs showing all necessary lines

E.g. I $2 \sin x \cos 60^\circ - 2 \cos x \sin 60^\circ = \cos x \cos 30^\circ + \sin x \sin 30^\circ$

$$\sin x (2 \cos 60^\circ - \sin 30^\circ) = \cos x (\cos 30^\circ + 2 \sin 60^\circ)$$

1. collect terms

$$(2 \cos 60^\circ - \sin 30^\circ) \tan x = \cos 30^\circ + 2 \sin 60^\circ$$

2. $\frac{\sin x}{\cos x} = \tan x$ so M1

$$\tan x = \frac{\cos 30^\circ + 2 \sin 60^\circ}{2 \cos 60^\circ - \sin 30^\circ} = \frac{\frac{\sqrt{3}}{2} + \sqrt{3}}{1 - \frac{1}{2}} = 3\sqrt{3}$$

3..uses values and completes proof A1*

E.g II

$$2 \sin x \times \frac{1}{2} - 2 \cos x \times \frac{\sqrt{3}}{2} = \cos x \times \frac{\sqrt{3}}{2} + \sin x \times \frac{1}{2}$$

1.uses values

$$\Rightarrow \frac{1}{2} \sin x = \frac{3\sqrt{3}}{2} \cos x$$

2.collects terms so M1

$$\Rightarrow \tan x = 3\sqrt{3}$$

3. $\frac{\sin x}{\cos x} = \tan x$ completes proof A1*

(b) Hence

B1: Deduces that $x = 2\theta + 60^\circ$ o.e such as $\theta = \frac{x - 60^\circ}{2}$

This is implied for sight of the equation $\tan(2\theta + 60^\circ) = 3\sqrt{3}$

M1: Proceeds from $\tan(2\theta \pm \alpha^\circ) = 3\sqrt{3} \Rightarrow 2\theta \pm \alpha^\circ = \text{one of } 79.1^\circ, 259.1^\circ, \dots$ where $\alpha \neq 0$

One angle for $\arctan(3\sqrt{3})$ must be correct in degrees or radians(3sf). FYI radian answers 1.38, 4.52

dM1: Correct method to find one value of θ from their $2\theta \pm \alpha^\circ = 79.1^\circ$ to $\theta = \frac{79.1^\circ \mp \alpha^\circ}{2}$

This is dependent upon one angle being correct, which must be in degrees, for $\arctan(3\sqrt{3})$

$$\tan(2\theta + 60^\circ) = 3\sqrt{3} \Rightarrow \theta = 9.6^\circ \text{ would imply B1 M1 dM1}$$

A1: $\theta = \text{awrt } 9.6^\circ, 99.6^\circ$ with no other values given in the range

Otherwise: Via the use of $\cos(2\theta + 30^\circ) = \cos 2\theta \cos 30^\circ - \sin 2\theta \sin 30^\circ$.

$$2 \sin 2\theta = \cos(2\theta + 30^\circ) \Rightarrow \tan 2\theta = \frac{\sqrt{3}}{5} \Rightarrow \theta = 9.6^\circ, 99.6^\circ$$

The order of the marks needs to match up to the main scheme so 0110 is possible.

B1: For achieving $\tan 2\theta = \frac{\sqrt{3}}{5}$ o.e so allow $\tan 2\theta = \text{awrt } 0.346$ or $\tan 2\theta = \frac{\cos 30^\circ}{2 + \sin 30^\circ}$

Or via double angle identities $\sqrt{3} \tan^2 \theta + 10 \tan \theta - \sqrt{3} = 0$ o.e.

M1: Attempts to use the compound angle identities to reach a form $\tan 2\theta = k$ where k is a constant not $3\sqrt{3}$ (or expression in trig terms such as $\cos 30$ etc as seen above)

Or via double angle identities reaches a 3TQ in $\tan \theta$

dM1: Correct order of operations from $\tan 2\theta = k$ leading to $\theta = \dots$

Correctly solves their $\sqrt{3} \tan^2 \theta + 10 \tan \theta - \sqrt{3} = 0$ leading to $\theta = \dots$

A1: $\theta = \text{awrt } 9.6^\circ, 99.6^\circ$ with no other values given in the range.

Note that $\tan(2\theta + 60^\circ) = 3\sqrt{3} \Rightarrow \theta = 9.6^\circ, 99.6^\circ$ is acceptable for full marks

Q14.

Question	Scheme	Marks	AOs
(a)	States or uses $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$	B1	1.2
	$\operatorname{cosec} \theta - \sin \theta = \frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta}$	M1	2.1
	$= \frac{\cos^2 \theta}{\sin \theta} = \cos \theta \times \frac{\cos \theta}{\sin \theta} = \cos \theta \cot \theta$ *	A1*	2.1
		(3)	
(b)	$\operatorname{cosec} x - \sin x = \cos x \cot(3x - 50^\circ)$ $\Rightarrow \cos x \cot x = \cos x \cot(3x - 50^\circ)$		
	$\cot x = \cot(3x - 50^\circ) \Rightarrow x = 3x - 50^\circ$	M1	3.1a
	$x = 25^\circ$	A1	1.1b
	Also $\cot x = \cot(3x - 50^\circ) \Rightarrow x + 180^\circ = 3x - 50^\circ$	M1	2.1
	$x = 115^\circ$	A1	1.1b
	Deduces $x = 90^\circ$	B1	2.2a
		(5)	
(8 marks)			
Notes:			

(a) Condone a full proof in x (or other variable) instead of θ 's here

B1: States or uses $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$ Do not accept $\operatorname{cosec} \theta = \frac{1}{\sin}$ with the θ missing

M1: For the key step in forming a single fraction/common denominator

E.g. $\operatorname{cosec} \theta - \sin \theta = \frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta}$. Allow if written separately $\frac{1}{\sin \theta} - \sin \theta = \frac{1}{\sin \theta} - \frac{\sin^2 \theta}{\sin \theta}$

Condone missing variables for this M mark

A1*: Shows careful work with all necessary steps shown leading to given answer. See scheme for necessary steps. There should not be any notational or bracketing errors.

(b) Condone θ 's instead of x 's here

M1: Uses part (a), cancels or factorises out the $\cos x$ term, to establish that one solution is found when $x = 3x - 50^\circ$.

You may see solutions where $\cot A - \cot B = 0 \Rightarrow \cot(A - B) = 0$ or $\tan A - \tan B = 0 \Rightarrow \tan(A - B) = 0$.

As long as they don't state $\cot A - \cot B = \cot(A - B)$ or $\tan A - \tan B = \tan(A - B)$ this is acceptable

A1: $x = 25^\circ$

M1: For the key step in realising that $\cot x$ has a period of 180° and a second solution can be found by solving $x + 180^\circ = 3x - 50^\circ$. The sight of $x = 115^\circ$ can imply this mark provided the step $x = 3x - 50^\circ$ has been seen. Using reciprocal functions it is for realising that $\tan x$ has a period of 180°

A1: $x = 115^\circ$ Withhold this mark if there are additional values in the range $(0, 180)$ but ignore values outside.

B1: Deduces that a solution can be found from $\cos x = 0 \Rightarrow x = 90^\circ$. Ignore additional values here.

.....
Solutions with limited working. The question demands that candidates show all stages of working.

SC: $\cos x \cot x = \cos x \cot(3x - 50^\circ) \Rightarrow \cot x = \cot(3x - 50^\circ) \Rightarrow x = 25^\circ, 115^\circ$

They have shown some working so can score B1, B1 marked on open as 11000

Alt 1- Right hand side to left hand side

Question	Scheme	Marks	AOs
(a)	States or uses $\cot \theta = \frac{\cos \theta}{\sin \theta}$	B1	1.2
	$\cos \theta \cot \theta = \frac{\cos^2 \theta}{\sin \theta} = \frac{1 - \sin^2 \theta}{\sin \theta}$	M1	2.1
	$= \frac{1}{\sin \theta} - \sin \theta = \operatorname{cosec} \theta - \sin \theta$ *	A1*	2.1
		(3)	

Alt 2- Works on both sides

Question	Scheme	Marks	AOs
(a)	States or uses $\cot \theta = \frac{\cos \theta}{\sin \theta}$ or $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$	B1	1.2
	$LHS = \frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta} = \frac{\cos^2 \theta}{\sin \theta}$ $RHS = \cos \theta \cot \theta = \frac{\cos^2 \theta}{\sin \theta}$	M1	2.1
	States a conclusion E.g. "HENCE TRUE", "QED" or $\operatorname{cosec} \theta - \sin \theta \equiv \cos \theta \cot \theta$ o.e. (condone = for $\equiv$)	A1*	2.1
		(3)	

Alt (b)

Question	Scheme	Marks	AOs
	$\cot x = \cot(3x - 50^\circ) \Rightarrow \frac{\cos x}{\sin x} = \frac{\cos(3x - 50^\circ)}{\sin(3x - 50^\circ)}$ $\sin(3x - 50^\circ) \cos x - \cos(3x - 50^\circ) \sin x = 0$ $\sin((3x - 50^\circ) - x) = 0$ $2x - 50^\circ = 0$	M1	3.1a
	$x = 25^\circ$	A1	1.1b
	Also $2x - 50^\circ = 180^\circ$	M1	2.1
	$x = 115^\circ$	A1	1.1b
	Deduces $\cos x = 0 \Rightarrow x = 90^\circ$	B1	2.2a
		(5)	

Q15.

Question	Scheme	Marks	AOs
(a)	Uses $\cos^2 x = 1 - \sin^2 x \Rightarrow 3 \sin^2 x + \sin x + 8 = 9(1 - \sin^2 x)$	M1	3.1a
	$\Rightarrow 12 \sin^2 x + \sin x - 1 = 0$	A1	1.1b
	$\Rightarrow (4 \sin x - 1)(3 \sin x + 1) = 0$	M1	1.1b
	$\Rightarrow \sin x = \frac{1}{4}, -\frac{1}{3}$	A1	1.1b
	Uses arcsin to obtain two correct values	M1	1.1b
	All four of $x = 14.48^\circ, 165.52^\circ, -19.47^\circ, -160.53^\circ$	A1	1.1b
		(6)	
(b)	Attempts $2\theta - 30^\circ = -19.47^\circ$	M1	3.1a
	$\Rightarrow \theta = 5.26^\circ$	A1ft	1.1b
		(2)	
(8 marks)			
Notes:			
(a) M1: Substitutes $\cos^2 x = 1 - \sin^2 x$ into $3 \sin^2 x + \sin x + 8 = 9 \cos^2 x$ to create a quadratic equation in just $\sin x$. A1: $12 \sin^2 x + \sin x - 1 = 0$ or exact equivalent M1: Attempts to solve their quadratic equation in $\sin x$ by a suitable method. These could include factorisation, formula or completing the square. A1: $\sin x = \frac{1}{4}, -\frac{1}{3}$ M1: Obtains two correct values for their $\sin x = k$ A1: All four of $x = 14.48^\circ, 165.52^\circ, -19.47^\circ, -160.53^\circ$ (b) M1: For setting $2\theta - 30^\circ = \text{their } -19.47^\circ$ A1ft: $\theta = 5.26^\circ$ but allow a follow through on their -19.47°			

Q16.

Question	Scheme	Marks	AOs	
	(i) $4\sin x = \sec x$, $0 \leq x < \frac{\pi}{2}$; (ii) $5\sin \theta - 5\cos \theta = 2$, $0 \leq \theta < 360^\circ$			
(i) Way 1	For $\sec x = \frac{1}{\cos x}$	B1	1.2	
	$\{4\sin x = \sec x \Rightarrow\} 4\sin x \cos x = 1 \Rightarrow 2\sin 2x = 1 \Rightarrow \sin 2x = \frac{1}{2}$	M1	3.1a	
	$x = \frac{1}{2}\arcsin\left(\frac{1}{2}\right)$ or $\frac{1}{2}\left(\pi - \arcsin\left(\frac{1}{2}\right)\right) \Rightarrow x = \frac{\pi}{12}, \frac{5\pi}{12}$	dM1	1.1b	
		A1	1.1b	
		(4)		
(i) Way 2	For $\sec x = \frac{1}{\cos x}$	B1	1.2	
	$\{4\sin x = \sec x \Rightarrow\} 4\sin x \cos x = 1 \Rightarrow 16\sin^2 x \cos^2 x = 1$ $16\sin^2 x(1 - \sin^2 x) = 1$ $16(1 - \cos^2 x)\cos^2 x = 1$ $16\sin^4 x - 16\sin^2 x + 1 = 0$ $16\cos^4 x - 16\cos^2 x + 1 = 0$ $\sin^2 x$ or $\cos^2 x = \frac{16 \pm \sqrt{192}}{32} \left\{ = \frac{2 \pm \sqrt{3}}{4} \text{ or } 0.933\dots, 0.066\dots \right\}$	M1	3.1a	
	$x = \arcsin\left(\sqrt{\frac{2 \pm \sqrt{3}}{4}}\right)$ or $x = \arccos\left(\sqrt{\frac{2 \pm \sqrt{3}}{4}}\right) \Rightarrow x = \frac{\pi}{12}, \frac{5\pi}{12}$	dM1	1.1b	
		A1	1.1b	
		(4)		
(ii)	Complete strategy, i.e. - Expresses $5\sin \theta - 5\cos \theta = 2$ in the form $R\sin(\theta - \alpha) = 2$, finds both R and α, and proceeds to $\sin(\theta - \alpha) = k$, $k < 1$, $k \neq 0$ - Applies $(5\sin \theta - 5\cos \theta)^2 = 2^2$, followed by applying both $\cos^2 \theta + \sin^2 \theta = 1$ and $\sin 2\theta = 2\sin \theta \cos \theta$ to proceed to $\sin 2\theta = k$, $k < 1$, $k \neq 0$	M1	3.1a	
	$R = \sqrt{50}$ $\tan \alpha = 1 \Rightarrow \alpha = 45^\circ$	$(5\sin \theta - 5\cos \theta)^2 = 2^2 \Rightarrow$ $25\sin^2 \theta + 25\cos^2 \theta - 50\sin \theta \cos \theta = 4$ $\Rightarrow 25 - 25\sin 2\theta = 4$	M1	1.1b
	$\sin(\theta - 45^\circ) = \frac{2}{\sqrt{50}}$	$\sin 2\theta = \frac{21}{25}$	A1	1.1b
	dependent on the first M mark			
	e.g. $\theta = \arcsin\left(\frac{2}{\sqrt{50}}\right) + 45^\circ$	e.g. $\theta = \frac{1}{2}\left(\arcsin\left(\frac{21}{25}\right)\right)$	dM1	1.1b
	$\theta = \text{awrt } 61.4^\circ, \text{ awrt } 208.6^\circ$		A1	2.1
	Note: Working in radians does not affect any of the first 4 marks			
			(5)	
(9 marks)				

Question	Scheme	Marks	AOs
	(ii) $5\sin\theta - 5\cos\theta = 2, 0 \leq \theta < 360^\circ$		
(ii) Alt 1	Complete strategy, i.e. • Attempts to apply $(5\sin\theta)^2 = (2+5\cos\theta)^2$ or $(5\sin\theta - 2)^2 = (5\cos\theta)^2$ followed by applying $\cos^2\theta + \sin^2\theta = 1$ and solving a quadratic equation in either $\sin\theta$ or $\cos\theta$ to give at least one of $\sin\theta = k$ or $\cos\theta = k, k < 1, k \neq 0$	M1	3.1a
	e.g. $25\sin^2\theta = 4 + 20\cos\theta + 25\cos^2\theta$ $\Rightarrow 25(1 - \cos^2\theta) = 4 + 20\cos\theta + 25\cos^2\theta$ or e.g. $25\sin^2\theta - 20\sin\theta + 4 = 25\cos^2\theta$ $\Rightarrow 25\sin^2\theta - 20\sin\theta + 4 = 25(1 - \sin^2\theta)$	M1	1.1b
	$50\cos^2\theta + 20\cos\theta - 21 = 0$ $50\sin^2\theta - 20\sin\theta - 21 = 0$		
	$\cos\theta = \frac{-20 \pm \sqrt{4600}}{100}, \text{ o.e.}$ $\sin\theta = \frac{20 \pm \sqrt{4600}}{100}, \text{ o.e.}$	A1	1.1b
	dependent on the first M mark e.g. $\theta = \arccos\left(\frac{-2 + \sqrt{46}}{10}\right)$ e.g. $\theta = \arcsin\left(\frac{2 + \sqrt{46}}{10}\right)$	dM1	1.1b
	$\theta = \text{awrt } 61.4^\circ, \text{ awrt } 208.6^\circ$	A1	2.1
		(5)	
Notes for Question			
(i)			
B1:	For recalling that $\sec x = \frac{1}{\cos x}$		
M1:	Correct strategy of • Way 1: applying $\sin 2x = 2\sin x \cos x$ and proceeding to $\sin 2x = k, k \leq 1, k \neq 0$ • Way 2: squaring both sides, applying $\cos^2 x + \sin^2 x = 1$ and solving a quadratic equation in either $\sin^2 x$ or $\cos^2 x$ to give $\sin^2 x = k$ or $\cos^2 x = k, k \leq 1, k \neq 0$		
dM1:	Uses the correct order of operations to find at least one value for x in either radians or degrees		
A1:	Clear reasoning to achieve both $x = \frac{\pi}{12}, \frac{5\pi}{12}$ and no other values in the range $0 \leq x < \frac{\pi}{2}$		
Note:	Give dM1 for $\sin 2x = \frac{1}{2} \Rightarrow$ any of $\frac{\pi}{12}, \frac{5\pi}{12}, 15^\circ, 75^\circ, \text{ awrt } 0.26 \text{ or awrt } 1.3$		
Note:	Give special case, SC B1M0M0A0 for writing down any of $\frac{\pi}{12}, \frac{5\pi}{12}, 15^\circ \text{ or } 75^\circ$ with no working		

Notes for Question Continued	
(ii)	
M1:	See scheme
Note:	Alternative strategy: Expresses $5\sin\theta - 5\cos\theta = 2$ in the form $R\cos(\theta + \alpha) = -2$, finds both R and α , and proceeds to $\cos(\theta + \alpha) = k$, $ k < 1$, $k \neq 0$
M1:	Either - uses $R\sin(\theta - \alpha)$ to find the values of both R and α - attempts to apply $(5\sin\theta - 5\cos\theta)^2 = 2^2$, uses $\cos^2\theta + \sin^2\theta = 1$ and proceeds to find an equation of the form $\pm\lambda \pm \mu\sin 2\theta = \pm\beta$ or $\pm\mu\sin 2\theta = \pm\beta$; $\mu \neq 0$ - attempts to apply $(5\sin\theta)^2 = (2 + 5\cos\theta)^2$ or $(5\sin\theta - 2)^2 = (5\cos\theta)^2$ and uses $\cos^2\theta + \sin^2\theta = 1$ to form an equation in $\cos\theta$ only or $\sin\theta$ only
A1:	For $\sin(\theta - 45^\circ) = \frac{2}{\sqrt{50}}$, o.e., $\cos(\theta + 45^\circ) = -\frac{2}{\sqrt{50}}$, o.e. or $\sin 2\theta = \frac{21}{25}$, o.e. or $\cos\theta = \frac{-20 \pm \sqrt{4600}}{100}$, o.e. or $\cos\theta = \text{awrt } 0.48$, awrt -0.88 or $\sin\theta = \frac{20 \pm \sqrt{4600}}{100}$, o.e., or $\sin\theta = \text{awrt } 0.88$, awrt -0.48
Note:	$\sin(\theta - 45^\circ)$, $\cos(\theta + 45^\circ)$, $\sin 2\theta$ must be made the subject for A1
dM1:	dependent on the first M mark Uses the correct order of operations to find at least one value for x in either degrees or radians
Note:	dM1 can also be given for $\theta = 180^\circ - \arcsin\left(\frac{2}{\sqrt{50}}\right) + 45^\circ$ or $\theta = \frac{1}{2}\left(180^\circ - \arcsin\left(\frac{21}{25}\right)\right)$
A1:	Clear reasoning to achieve both $\theta = \text{awrt } 61.4^\circ$, awrt 208.6° and no other values in the range $0 \leq \theta < 360^\circ$
Note:	Give M0M0A0M0A0 for writing down any of $\theta = \text{awrt } 61.4^\circ$, awrt 208.6° with no working
Note:	<u>Alternative solutions:</u> (to be marked in the same way as Alt 1): $5\sin\theta - 5\cos\theta = 2 \Rightarrow 5\tan\theta - 5 = 2\sec\theta \Rightarrow (5\tan\theta - 5)^2 = (2\sec\theta)^2$ $\Rightarrow 25\tan^2\theta - 50\tan\theta + 25 = 4\sec^2\theta \Rightarrow 25\tan^2\theta - 50\tan\theta + 25 = 4(1 + \tan^2\theta)$ $\Rightarrow 21\tan^2\theta - 50\tan\theta + 21 = 0 \Rightarrow \tan\theta = \frac{50 \pm \sqrt{736}}{42} = \frac{25 \pm 2\sqrt{46}}{21} = 1.8364\dots, 0.5445\dots$ $\Rightarrow \theta = \text{awrt } 61.4^\circ$, awrt 208.6° only $5\sin\theta - 5\cos\theta = 2 \Rightarrow 5 - 5\cot\theta = 2\text{cosec}\theta \Rightarrow (5 - 5\cot\theta)^2 = (2\text{cosec}\theta)^2$ $\Rightarrow 25 - 50\cot\theta + 25\cot^2\theta = 4\text{cosec}^2\theta \Rightarrow 25 - 50\cot\theta + 25\cot^2\theta = 4(1 + \cot^2\theta)$ $\Rightarrow 21\cot^2\theta - 50\cot\theta + 21 = 0 \Rightarrow \cot\theta = \frac{50 \pm \sqrt{736}}{42} = \frac{25 \pm 2\sqrt{46}}{21} = 1.8364\dots, 0.5445\dots$ $\Rightarrow \theta = \text{awrt } 61.4^\circ$, awrt 208.6° only

Q17.

Question	Scheme	Marks	AOs
(a)	$x^2 \tan y = 9 \Rightarrow 2x \tan y + x^2 \sec^2 y \frac{dy}{dx} = 0$	M1 A1	3.1a 1.1b
	Full method to get $\frac{dy}{dx}$ in terms of x using $\sec^2 y = 1 + \tan^2 y = 1 + f(x)$	M1	1.1b
	$\frac{dy}{dx} = \frac{-2x \times \frac{9}{x^2}}{x^2 \left(1 + \frac{81}{x^4}\right)} = \frac{-18x}{x^4 + 81} \quad *$	A1*	2.1
		(4)	
(b)	$\frac{dy}{dx} = \frac{-18x}{x^4 + 81}$ $\Rightarrow \frac{d^2y}{dx^2} = \frac{-18 \times (x^4 + 81) - (-18x)(4x^3)}{(x^4 + 81)^2} = \frac{54(x^4 - 27)}{(x^4 + 81)^2} \text{ o.e.}$	M1 A1	1.1b 1.1b
	States that when $x < \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} < 0$ when $x = \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} = 0$ AND when $x > \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} > 0$ giving a point of inflection when $x = \sqrt[4]{27}$	A1	2.4
		(3)	
		(7 marks)	
Notes:			

(a)

M1: Attempts to differentiate $\tan y$ implicitly. Eg. $\tan y \rightarrow \sec^2 y \frac{dy}{dx}$ or $\cot y \rightarrow -\operatorname{cosec}^2 y \frac{dy}{dx}$

You may well see an attempt $\tan y = \frac{9}{x^2} \Rightarrow \sec^2 y \frac{dy}{dx} = \dots$

When a candidate writes $x^2 \tan y = 9 \Rightarrow x = 3 \tan^{\frac{1}{2}} y$ the mark is scored for $\tan^{\frac{1}{2}} y \rightarrow \dots \tan^{\frac{3}{2}} y \sec^2 y$

A1: Correct differentiation $2x \tan y + x^2 \sec^2 y \frac{dy}{dx} = 0$

Allow also $\sec^2 y \frac{dy}{dx} = -\frac{18}{x^3}$ or $2x = -9 \operatorname{cosec}^2 y \frac{dy}{dx}$ amongst others

M1: Full method to get $\frac{dy}{dx}$ in terms of x using $\sec^2 y = 1 + \tan^2 y = 1 + f(x)$

A1*: Proceeds correctly to the given answer of $\frac{dy}{dx} = \frac{-18x}{x^4 + 81}$

(b)

M1: Attempts to differentiate the given expression using the product or quotient rule.

For example look for a correct attempt at $\frac{vu' - uv'}{v^2}$ with $u = -18x, v = x^4 + 81, u' = \pm 18, v' = \dots x^3$

If no method is seen or implied award for $\frac{\pm 18 \times (x^4 + 81) \pm 18x(x^3)}{(x^4 + 81)^2}$

Using the product rule award for $\pm 18(x^4 + 81)^{-1} \pm 18x(x^4 + 81)^{-2} \times cx^3$

A1: Correct simplified $\frac{d^2y}{dx^2} = \frac{54(x^4 - 27)}{(x^4 + 81)^2}$ o.e. such as $\frac{d^2y}{dx^2} = \frac{54x^4 - 1458}{(x^4 + 81)^2}$

Alternatively score for showing that when a correct (unsimplified) $\frac{d^2y}{dx^2} = 0 \Rightarrow x^4 = 27 \Rightarrow x = \sqrt[4]{27}$

Or for substituting $x = \sqrt[4]{27}$ into an unsimplified but correct $\frac{d^2y}{dx^2}$ and showing that it is 0

A1: Correct explanation with a minimal conclusion and correct second derivative.

See scheme.

It can be also be argued from $x^4 < 27$, $x^4 = 27$ and $x^4 > 27$ provided the conclusion states that the point of inflection is at $x = \sqrt[4]{27}$

Alternatively substitutes values of x either side of $\sqrt[4]{27}$ and at $\sqrt[4]{27}$, into $\frac{d^3y}{dx^3}$, finds all three values and makes a minimal conclusion.

A different method involves finding $\frac{d^3y}{dx^3}$ and showing that $\frac{d^3y}{dx^3} \neq 0$ and $\frac{d^2y}{dx^2} = 0$ when $x = \sqrt[4]{27}$

$$\text{FYI } \frac{d^3y}{dx^3} = \frac{23328x^3}{(x^4 + 81)^3} = 0.219 \text{ when } x = \sqrt[4]{27}$$

Alternative part (a) using arctan

M1: Sets $y = \arctan \frac{9}{x^2} \rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{9}{x^2}\right)^2} \times \dots$ where ... could be 1

A2: $y = \arctan \frac{9}{x^2} \rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{9}{x^2}\right)^2} \times -\frac{18}{x^3}$

A1*: $\frac{dy}{dx} = \frac{1}{1 + \frac{81}{x^4}} \times -\frac{18}{x^3} = \frac{-18x}{x^4 + 1}$ showing correct intermediate step and no errors.

Q18.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{\cos 3\theta}{\sin \theta} + \frac{\sin 3\theta}{\cos \theta} = \frac{\cos 3\theta \cos \theta + \sin 3\theta \sin \theta}{\sin \theta \cos \theta}$	M1	This mark is given for a method to form a single fraction
	$= \frac{\cos(3\theta - \theta)}{\sin \theta \cos \theta}$	M1	This mark is given for a method to use a compound angle formula on the numerator
	$= \frac{\cos 2\theta}{\frac{1}{2} \sin 2\theta}$	M1	This mark is given for a method to use a compound angle formula on the denominator
	$= 2 \cot 2\theta$	A1	This mark is given for a fully correct proof to show the answer required
(b)	$\tan 2\theta = \frac{1}{2}$	M1	This mark is given for deducing that the value of $\tan 2\theta$
	$180^\circ + 26.6^\circ$	M1	This mark is given for finding the solution in the third quadrant for $\arctan \frac{1}{2}$
	$\theta = 103.3^\circ$	A1	This mark is given for finding a correct value for θ
(Total 7 marks)			

Q19.

Question	Scheme	Marks	AOs
(a)	$\frac{1 - \cos 2\theta + \sin 2\theta}{1 + \cos 2\theta + \sin 2\theta} = \frac{1 - (1 - 2\sin^2 \theta) + 2\sin \theta \cos \theta}{1 + \cos 2\theta + \sin 2\theta}$	M1	2.1
	or $\frac{1 - \cos 2\theta + \sin 2\theta}{1 + \cos 2\theta + \sin 2\theta} = \frac{1 - \cos 2\theta + \sin 2\theta}{1 + (2\cos^2 \theta - 1) + 2\sin \theta \cos \theta}$		
	$\frac{1 - \cos 2\theta + \sin 2\theta}{1 + \cos 2\theta + \sin 2\theta} = \frac{1 - (1 - 2\sin^2 \theta) + 2\sin \theta \cos \theta}{1 + (2\cos^2 \theta - 1) + 2\sin \theta \cos \theta}$	A1	1.1b
	$= \frac{2\sin^2 \theta + 2\sin \theta \cos \theta}{2\cos^2 \theta + 2\sin \theta \cos \theta} = \frac{2\sin \theta (\sin \theta + \cos \theta)}{2\cos \theta (\cos \theta + \sin \theta)}$	dM1	2.1
	$= \frac{\sin \theta}{\cos \theta} = \tan \theta^*$	A1*	1.1b
		(4)	
(b)	$\frac{1 - \cos 4x + \sin 4x}{1 + \cos 4x + \sin 4x} = 3 \sin 2x \Rightarrow \tan 2x = 3 \sin 2x \quad \text{o.e.}$	M1	3.1a
	$\Rightarrow \sin 2x - 3 \sin 2x \cos 2x = 0$ $\Rightarrow \sin 2x (1 - 3 \cos 2x) = 0$ $\Rightarrow (\sin 2x = 0,) \cos 2x = \frac{1}{3}$	A1	1.1b
	$x = 90^\circ, \text{ awrt } 35.3^\circ, \text{ awrt } 144.7^\circ$	A1 A1	1.1b 2.1
		(4)	
(8 marks)			
Notes			

(a)

M1: Attempts to use a correct double angle formulae for both $\sin 2\theta$ and $\cos 2\theta$ (seen once).

The application of the formula for $\cos 2\theta$ must be the one that cancels out the "1"

So look for $\cos 2\theta = 1 - 2\sin^2\theta$ in the numerator or $\cos 2\theta = 2\cos^2\theta - 1$ in the denominator

Note that $\cos 2\theta = \cos^2\theta - \sin^2\theta$ may be used as well as using $\cos^2\theta + \sin^2\theta = 1$

A1: $\frac{1 - (1 - 2\sin^2\theta) + 2\sin\theta\cos\theta}{1 + (2\cos^2\theta - 1) + 2\sin\theta\cos\theta}$ or $\frac{2\sin^2\theta + 2\sin\theta\cos\theta}{2\cos^2\theta + 2\sin\theta\cos\theta}$

dM1: Factorises numerator and denominator in order to demonstrate cancelling of $(\sin\theta + \cos\theta)$

A1*: Fully correct proof with no errors.

You must see an intermediate line of $\frac{2\sin\theta(\sin\theta + \cos\theta)}{2\cos\theta(\cos\theta + \sin\theta)}$ or $\frac{\sin\theta}{\cos\theta}$ or even $\frac{2\sin\theta}{2\cos\theta}$

Withhold this mark if you see, within the body of the proof,

- notational errors. E.g. $\cos 2\theta = 1 - 2\sin^2$ or $\cos\theta^2$ for $\cos^2\theta$
- mixed variables. E.g. $\cos 2\theta = 2\cos^2x - 1$

(b)

M1: Makes the connection with part (a) and writes the lhs as $\tan 2x$. Condone $x \leftrightarrow \theta$ $\tan 2\theta = 3 \sin 2\theta$

A1: Obtains $\cos 2x = \frac{1}{3}$ o.e. with $x \leftrightarrow \theta$. You may see $\sin^2 x = \frac{1}{3}$ or $\cos^2 x = \frac{2}{3}$ after use of double angle formulae.

A1: Two "correct" values. Condone accuracy of awrt 90° , 35° , 145°

Also condone radian values here. Look for 2 of awrt 0.62, 1.57, 2.53

A1: All correct (allow awrt) and no other values in range. Condone $x \leftrightarrow \theta$ if used consistently

.....
Answers without working in (b): Just answers and no working score 0 marks.

If the first line is written out, i.e. $\tan 2x = 3 \sin 2x$ followed by all three correct answers score 1100.

Q20.

Question	Scheme	Marks	AOs
(a)	$R = \sqrt{5}$	B1	1.1b
	$\tan \alpha = \frac{1}{2}$ or $\sin \alpha = \frac{1}{\sqrt{5}}$ or $\cos \alpha = \frac{2}{\sqrt{5}} \Rightarrow \alpha = \dots$	M1	1.1b
	$\alpha = 0.464$	A1	1.1b
		(3)	
(b)(i)	$3 + 2\sqrt{5}$	B1ft	3.4
(ii)	$\cos(0.5t + 0.464) = 1 \Rightarrow 0.5t + 0.464 = 2\pi$ $\Rightarrow t = \dots$	M1	3.4
	$t = 11.6$	A1	1.1b
		(3)	
(c)	$3 + 2\sqrt{5} \cos(0.5t + 0.464) = 0$ $\cos(0.5t + 0.464) = -\frac{3}{2\sqrt{5}}$	M1	3.4
	$\cos(0.5t + 0.464) = -\frac{3}{2\sqrt{5}} \Rightarrow 0.5t + 0.464 = \cos^{-1}\left(-\frac{3}{2\sqrt{5}}\right)$ $\Rightarrow t = 2\left(\cos^{-1}\left(-\frac{3}{2\sqrt{5}}\right) - 0.464\right)$	dM1	1.1b
	So the time required is e.g.: $2(3.977\dots - 0.464) - 2(2.306\dots - 0.464)$	dM1	3.1b
	$= 3.34$	A1	1.1b
		(4)	
(d)	e.g. the "3" would need to vary	B1	3.5c
		(1)	
(11 marks)			

Notes
(a) B1: $R = \sqrt{5}$ only. M1: Proceeds to a value for α from $\tan \alpha = \pm \frac{1}{2}$ or $\sin \alpha = \pm \frac{1}{\sqrt{5}}$ or $\cos \alpha = \pm \frac{2}{\sqrt{5}}$ It is implied by either awrt 0.464 (radians) or awrt 26.6 (degrees) A1: $\alpha = \text{awrt } 0.464$ (b)(i) B1ft: For $(3 + 2\sqrt{5})$ m or awrt 7.47 m and remember to isw. Condone lack of units. Follow through on their R value so allow $3 + 2 \times \text{Their } R$. (Allow in decimals with at least 3sf accuracy) (b)(ii) M1: Uses $0.5t \pm "0.464" = 2\pi$ to obtain a value for t Follow through on their 0.464 but this angle must be in radians. It is possible in degrees but only using $0.5t \pm "26.6" = 360$ A1: Awrt 11.6

Alternative for (b):

$$H = 3 + 4 \cos(0.5t) - 2 \sin(0.5t) \Rightarrow \frac{dH}{dt} = -2 \sin(0.5t) - \cos(0.5t) = 0$$

$$\Rightarrow \tan(0.5t) = -\frac{1}{2} \Rightarrow 0.5t = 2.677..., 5.819... \Rightarrow t = 5.36, 11.6$$

$$t = 11.6 \Rightarrow H = 7.47$$

Score as follows:

M1: For a complete method:

Attempts $\frac{dH}{dt}$ and attempts to solve $\frac{dH}{dt} = 0$ for t

A1: For $t = \text{awrt } 11.6$

B1ft: For awrt 7.47 or $3 + 2 \times \text{Their } R$

(c)

M1: Uses the model and sets $3 + 2\sqrt{5} \cos(\dots) = 0$ and proceeds to $\cos(\dots) = k$ where $|k| < 1$.

Allow e.g. $3 + 2\sqrt{5} \cos(\dots) < 0$

dM1: Solves $\cos(0.5t \pm 0.464) = k$ where $|k| < 1$ to obtain at least one value for t

This requires e.g. $2\left(\pi + \cos^{-1}(k) \pm \tan^{-1}\left(\frac{1}{2}\right)\right)$ or e.g. $2\left(\pi - \cos^{-1}(k) \pm \tan^{-1}\left(\frac{1}{2}\right)\right)$

Depends on the previous method mark.

dM1: A fully correct strategy to find the required duration. E.g. finds 2 consecutive values of t when $H = 0$ and subtracts. Alternatively finds t when H is minimum and uses the times found correctly to find the required duration.

Depends on the previous method mark.

Examples:

Second time at water level – first time at water level:

$$2\left(\pi + \cos^{-1}\left(\frac{3}{2\sqrt{5}}\right) - \tan^{-1}\left(\frac{1}{2}\right)\right) - 2\left(\pi - \cos^{-1}\left(\frac{3}{2\sqrt{5}}\right) - \tan^{-1}\left(\frac{1}{2}\right)\right) = 7.02685... - 3.68492...$$

$2 \times (\text{first time at minimum point} - \text{first time at water level}):$

$$2\left(2\left(\pi - \tan^{-1}\left(\frac{1}{2}\right)\right) - 2\left(\pi - \cos^{-1}\left(\frac{3}{2\sqrt{5}}\right) - \tan^{-1}\left(\frac{1}{2}\right)\right)\right) = 2(5.35589... - 3.68492...)$$

Note that both of these examples equate to $4\cos^{-1}\left(\frac{3}{2\sqrt{5}}\right)$ which is not immediately obvious

but may be seen as an overall method.

There may be other methods – if you are not sure if they deserve credit send to review.

A1: Correct value. Must be 3.34 (not awrt).

Special Cases in (c):

Note that if candidates have an incorrect α and have e.g. $3 + 2\sqrt{5} \cos(0.5t - 0.464)$, this has no impact on the final answer. So for candidates using $3 + 2\sqrt{5} \cos(0.5t \pm \alpha)$ in (c) allow all the marks including the A mark as a correct method should always lead to 3.34

Some values to look for:

$$0.5t \pm 0.464 = \pm 2.306, \pm 3.977, \pm 5.598, \pm 7.266$$

(d)

B1: Correct refinement e.g. As in scheme. If they suggest a specific function to replace the “3” then it must be sensible e.g. a trigonometric function rather than e.g. a quadratic/linear one.